

The idea of the Jacobi method is to try to zero the off-diagonal elements by a series of plane rotations. Accordingly, to set $a'_{pq} = 0$, equation (11.1.7) gives the following expression for the rotation angle ϕ :

$$\theta \equiv \cot 2\phi \equiv \frac{c^2 - s^2}{2sc} = \frac{a_{qq} - a_{pp}}{2a_{pq}} \quad (11.1.8)$$

If we let $t \equiv s/c$, the definition of θ can be rewritten

$$t^2 + 2t\theta - 1 = 0 \quad (11.1.9)$$

The smaller root of this equation corresponds to a rotation angle less than $\pi/4$ in magnitude; this choice at each stage gives the most stable reduction. Using the form of the quadratic formula with the discriminant in the denominator, we can write this smaller root as

$$t = \frac{\text{sgn}(\theta)}{|\theta| + \sqrt{\theta^2 + 1}} \quad (11.1.10)$$

If θ is so large that θ^2 would overflow on the computer, we set $t = 1/(2\theta)$. It now follows that

$$c = \frac{1}{\sqrt{t^2 + 1}} \quad (11.1.11)$$

$$s = tc \quad (11.1.12)$$

When we actually use equations (11.1.4) – (11.1.7) numerically, we rewrite them to minimize roundoff error. Equation (11.1.7) is replaced by

$$a'_{pq} = 0 \quad (11.1.13)$$

The idea in the remaining equations is to set the new quantity equal to the old quantity plus a small correction. Thus we can use (11.1.7) and (11.1.13) to eliminate a_{qq} from (11.1.5), giving

$$a'_{pp} = a_{pp} - ta_{pq} \quad (11.1.14)$$

Similarly,

$$a'_{qq} = a_{qq} + ta_{pq} \quad (11.1.15)$$

$$a'_{rp} = a_{rp} - s(a_{rq} + \tau a_{rp}) \quad (11.1.16)$$

$$a'_{rq} = a_{rq} + s(a_{rp} - \tau a_{rq}) \quad (11.1.17)$$

where $\tau (= \tan \phi/2)$ is defined by

$$\tau \equiv \frac{s}{1 + c} \quad (11.1.18)$$

One can see the convergence of the Jacobi method by considering the sum of the squares of the off-diagonal elements

$$S = \sum_{r \neq s} |a_{rs}|^2 \quad (11.1.19)$$